

Principal Component Analysis & Course Synthesis

INFO 521 · student notes · Module 7 · ~7 min read

The last module asks the other unsupervised question: not which *groups* the patients form, but which *directions* among the features actually matter. It ends by resolving all seven modules into one arc.

PCA from scratch

Six NHANES features are not six independent pieces of information — BMI and waist track each other at $r \approx 0.90$. PCA finds new axes (**principal components**) that are uncorrelated and ordered by how much variance each carries. The recipe: **center** each feature (else the first component points at the mean), **standardize** (variance is unit-squared — Module 6's scaling lesson again), form the sample covariance $\mathbf{C} = \frac{1}{N-1} \mathbf{X}^\top \mathbf{X}$, and take its **eigendecomposition**. Eigenvectors are the new axes; eigenvalues are the variance each carries.

Two views, one solution

PCA solves two problems that turn out identical. **Maximum variance**: the unit direction with the most projected spread is the top eigenvector. **Minimum reconstruction error**: projecting onto K directions and rebuilding with least squared error picks the same top- K eigenvectors. They coincide because variance kept + reconstruction error = total variance (fixed), so maximizing one minimizes the other. And that reconstruction picture is least squares — Module 1's projection geometry, now in feature space. **Least squares closes the course as it opened it.**

On the (BMI, waist) plane, one direction carries $\approx 95\%$ of the spread. On all six features the scree plot reads 37.3 / 21.8 / 17.2 / 12.5 / 9.7 / **1.3%** — no cliff after PC1 (clinical features are genuinely multi-dimensional), but a near-zero floor at PC6 announcing the known redundancy. The loadings are interpretable: **PC1** is an adiposity/metabolic axis (BMI + waist + HbA1c, HDL opposing), **PC2** an age-cholesterol axis.

Choosing K and honest limits

Choose the number of components by the **scree elbow**, a **variance threshold** ("keep 90%" $\rightarrow K = 5$ here), or — if the scores feed a supervised model — by **cross-validation** on that model (Module 2 closing the loop). But: PCA is **linear** (blind to curved structure); **scale-dependent** (standardize or units choose the components); interpretability is a gift, not a guarantee; and most importantly, **variance is not usefulness** — PCA never saw the label, so the direction that predicts hypertension may hide in a low-variance component you were about to discard. It is Module 6's cluster-versus-label caution, transposed to axes.

The PCA explorer (systolic vs diastolic BP) lets you rotate a candidate axis and watch variance-captured rise while reconstruction-error falls, both hitting their best at the same angle — the two views as one solution — with **Snap to PC1** to confirm.

The course in one arc

One dataset, five ways of seeing, each module's answer becoming the next one's question:

Move	Object
Loss (M1–M2)	$\hat{\mathbf{w}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$; flexibility controlled
Likelihood (M3)	same $\hat{\mathbf{w}} + \hat{\sigma}^2 + \text{Cov}[\hat{\mathbf{w}}]$
Posterior (M4)	$\mathcal{N}(\mathbf{m}_N, \mathbf{S}_N)$ — Part 1's endpoint
Approximation (M5)	Laplace $\mathbf{A}^{-1}$; Metropolis draws
Discovery (M6–M7)	clusters; principal axes — described, not diagnosed

And one object followed the whole way: $\mathbf{X}^\top \mathbf{X}$ — the loss bowl (M1), the likelihood curvature and Fisher information (M3), the posterior precision (M4), and the covariance whose eigenvectors PCA reads (M7). Same matrix, the whole course through.

What to remember

- PCA = eigendecomposition of the (standardized) covariance; eigenvalues are variance explained.
- Maximum variance and minimum reconstruction error are the same solution (least squares in feature space).
- Choose K by elbow, threshold, or downstream CV — and remember variance $\neq$ usefulness.
- The whole course is one dataset seen five ways, around the recurring matrix $\mathbf{X}^\top \mathbf{X}$.